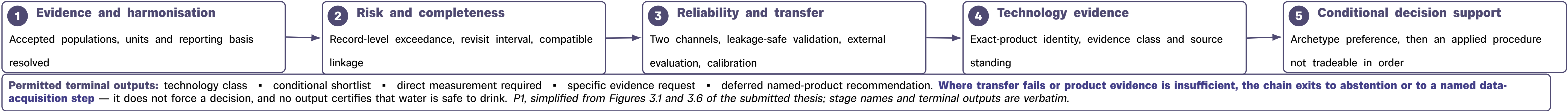


Predicting Drinking-Water Contamination Risk and Ranking Monitoring Technologies Under Uncertainty: An Evidence-Aware, Leakage-Safe Decision-Support Framework

Muhammad Faseeh Memon, CID 06058747
Supervisor: Professor Michael Templeton
Department of Civil and Environmental Engineering



Two decisions, one disconnected evidence base

A monitoring programme must answer two linked questions: *where should measurement effort go*, and *what should make the measurement*. What has been measured, what a model can be trusted to predict and what an instrument can be relied upon to detect are rarely joined.

Scope. India is the principal empirical application. Kenya and Malawi are bounded local comparators and support no national estimate; Bangladesh and Ghana appear only as literature-grounded applications of the decision logic, with no model transferred to either.

What the records could support. Six India parameters reached an accepted analytical population. Turbidity and *E. coli* reached none, so no baseline was claimed for either. Of 145 state-by-parameter cells with recorded cadence, none is revisited more often than annually at its median — a monitoring-design limit no model can repair.

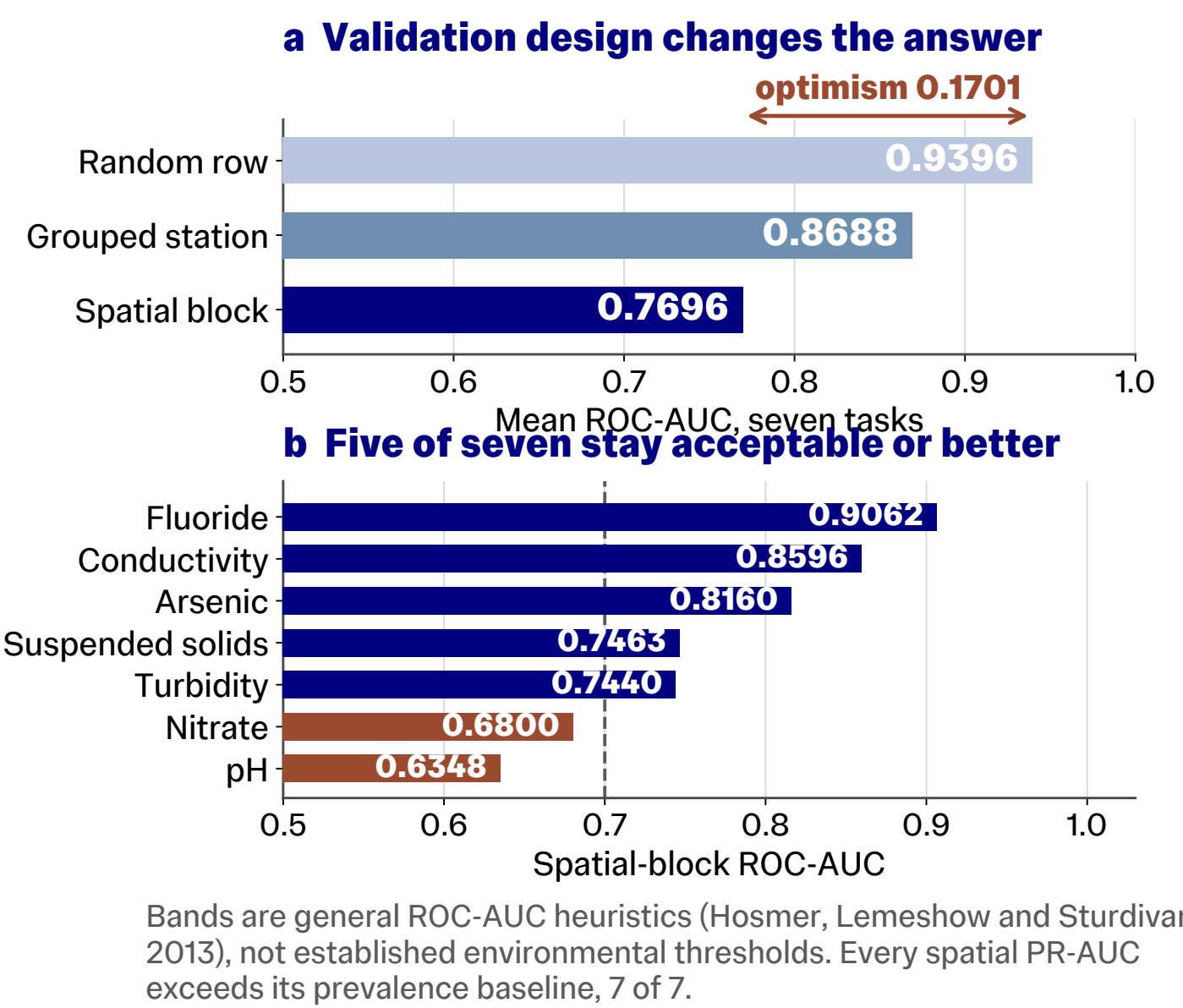
A recommendation is only as defensible as the evidence layer beneath it.

Hosmer, D., Lemeshow, S. & Sturdivant, R. (2013). *Applied logistic regression*. 3rd edn. Hoboken: Wiley. Hu, X., Dai, M., Sun, J. & Sunderland, E. (2023). *Current Environmental Health Reports*.

Every figure and number on this poster is traced to a page of the submitted thesis in *poster_source_traceability.md*.

Random splitting overstated performance

Seven tasks, three validation designs. Geographic blocking is what a programme extending into an unmonitored district should plan against.



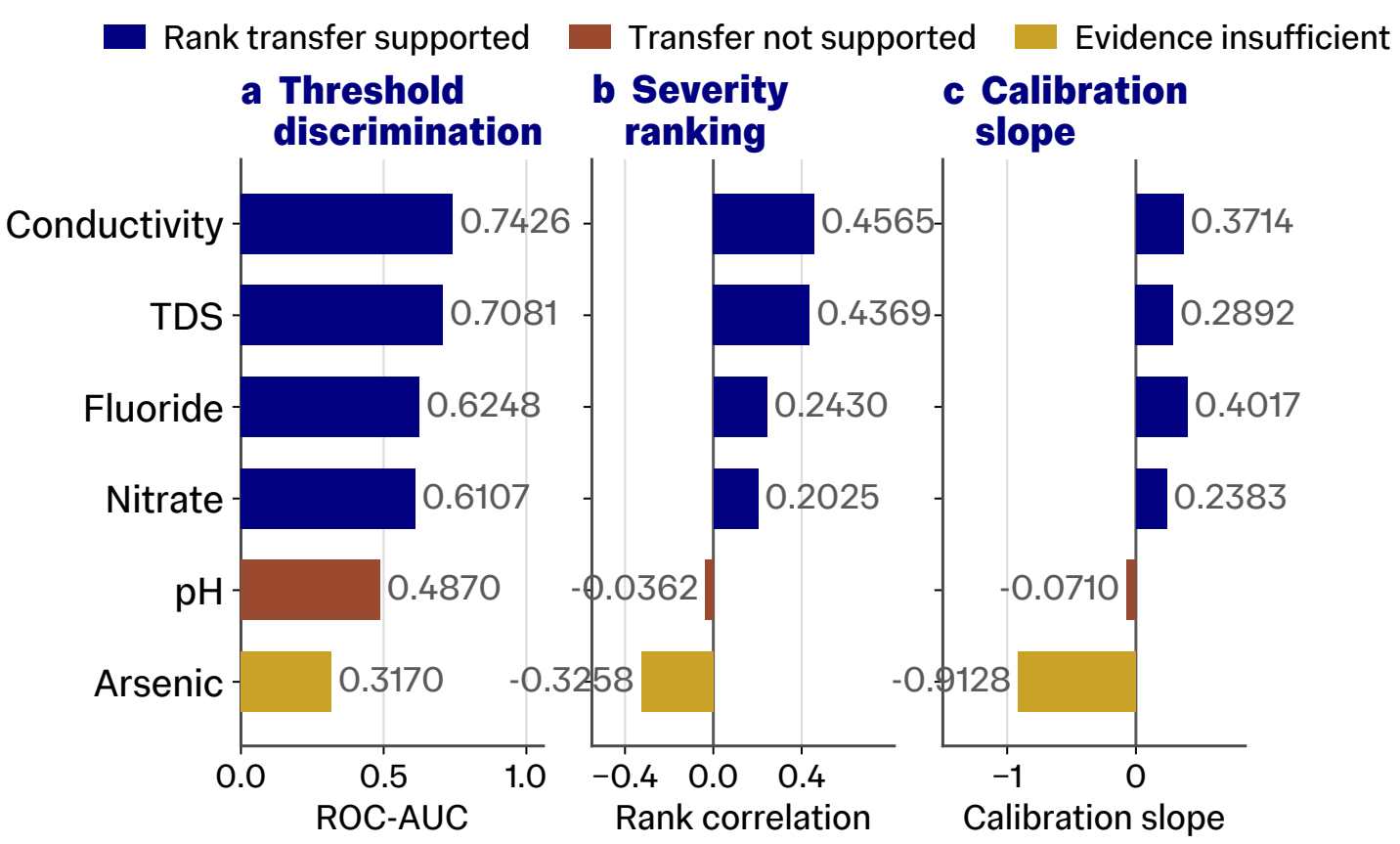
P2. Values from Table 4.3 and Section 4.3 of the submitted thesis.

The 0.1701 optimism decomposes into **0.0708** removed by grouping repeat visits to a station, and a further **0.0993** revealed only by geographic separation.

A scoped label-shuffle control collapsed spatial performance to **0.501** (95% interval 0.482 to 0.520): recoverable signal, not a label-independent spatial artefact.

Same-site history (Model B) was higher on all six shared tasks — a *random-row channel comparison*, not *geographic generalisation*.

Ranking transferred where calibration did not



A supported rank is not a calibrated probability: every India row carries negative Brier skill, so recalibration on local labels is required before any score is read as a local exceedance probability.

P3. Independent CGWB archives; values from Table 4.6 of the submitted thesis.

Conductivity, TDS, fluoride, nitrate — rank support, 4 of 6: sequence confirmatory sampling. **pH** — not supported in this evaluation: measure directly. **Arsenic** — evidence-insufficient on 119 observations at 116 sites. **Any probability use** — recalibrate on local labels first.

2.01× 2.40×

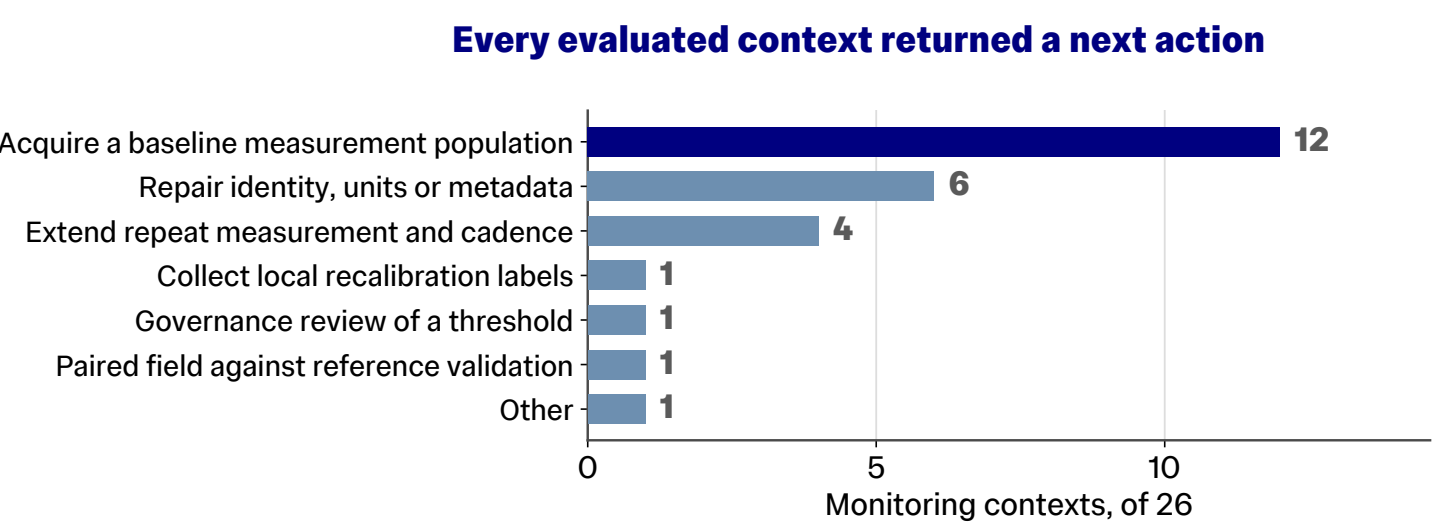
Confirmatory-test enrichment over the untargeted expectation at the **recorded operating point**: conductivity, 5,966 tests finding 2,838 exceeding sites against 1,410; TDS, 265 tests finding 154 against 64.2. Not usable for the other four parameters.

Rajasthan, worked through. 12,349 conductivity records at 1,641 sites, 54.06% above the 1,500 $\mu\text{S}/\text{cm}$ project convention — a record-level signal, not population exposure. Rank transfer sequences the sampling; the calibration slopes forbid reading any score as a local probability; four products carry stated capability and none passes the evidence gate. Terminal decision: a conditional class shortlist, product-level abstention.

Capability was common; product evidence was not

0 of 23

exact commercial products cleared the eleven-criterion evidence gate. Of **1,242** product–criterion cells (23 products × 54 criteria), **223** carry an exact-product document and only **9** carry independent evidence. The 33-entry landscape is separate: 23 products, 8 reference methods, 2 emerging classes. *This is abstention, not poor performance — missing documentation remains unknown evidence.*



P4. Direct counts across 26 parameter-context decisions in India, Kenya and Malawi, from Figure 4.14 of the submitted thesis. Not a score, and not 26 countries or sites.

What resource-constrained monitoring still needs. Neither cost nor energy per valid result can be derived for any of the 23 products, so neither saving can be priced. Threshold-near accuracy, operator effect, invalid tests, reliability, reagent stability and persistent site, sample, time and method identifiers are the recurring gaps. Of sixteen research priorities, only two concern sensor design.

Ranking can stay useful after calibration stops.

Where prediction or product evidence is insufficient, the framework returns a measurement or a named evidence request rather than manufacturing certainty.